1. NIH Chest X – rays:

This is one of the most famous and widely utilized large-scale medical datasets in the world, compiled to help train AI to detect pulmonary diseases.

Features	Details
Type of data	2D Chest X – ray
Number of images	Around 1.12 lakh front view X – rays from around 31 thousand patients
Available Labels	Around 15 (14 for different types of diseases/conditions and one for a healthy lung)
Dataset imbalance	Extremely high dataset imbalance
Observed Challenges	• There are multiple images per patient. Hence, the patient should not be put in both, testing and training phase. • A single patient can have multiple overlapping diseases.

The NIH Chest X-ray dataset is a big benchmark in medical computer vision. It is useful to create a sophisticated, complex model which must detect varying, concurrent diseases. While its massive size is excellent for deep learning, the inherent noise means that models trained on it often struggle to achieve perfect clinical accuracy without secondary human verification.

2. Diabetic Retinopathy Detection (EyePACS):

Diabetic retinopathy is a leading cause of blindness. This dataset was created to automate the grading of the disease from retinal photographs.

Feature	Details
Type of imaging data	High resolution photography of retina
Number of images	Around 35,000 training images
Available classes	5 classes: from 0 (no diabetic retinopathy) to 4 (proliferative diabetic retinopathy)
Dataset imbalance	High dataset imbalance found
Observed Challenges	• Many images are blurred or more/less exposed. There may even be dust on the lens when the image is taken. • The difference between the levels of diabetic retinopathy is very subtle, and the new symptoms are just a few pixels wide.

The EyePACS Diabetic Retinopathy dataset presents a highly realistic and challenging clinical scenario. It is a good dataset for learning about image preprocessing techniques. To succeed here, researchers typically have to crop the dark borders out of the images, apply custom color normalization (to make the blood vessels and lesions pop), and heavily augment the minority classes to prevent the AI from simply guessing "No DR" every single time.